

■ 3.9 Numerical Operations on Functions

■ 3.9.1 Numerical Mathematics in *Mathematica*

One of the most important features of *Mathematica* is its ability to give you exact, symbolic, results for computations. There are, however, computations where it is just mathematically impossible to get exact “closed form” results. In such cases, you can still often get approximate numerical results.

There is no “closed form” result for $\int_0^1 \sin(\exp(x)) dx$. *Mathematica* returns the integral in its original symbolic form.

```
In[1]:= Integrate[Sin[Exp[x]], {x, 0, 1}]
Out[1]= Integrate[Sin[E^x], {x, 0, 1}]
```

You can now take the symbolic form of the integral, and ask for its approximate numerical value.

```
In[2]:= N[%]
Out[2]= 0.874957
```

When *Mathematica* cannot find an explicit result for something like a definite integral, it returns a symbolic form. You can take this symbolic form, and try to get an approximate numerical value by applying **N**.

By giving a second argument to **N**, you can specify the numerical precision to use.

```
In[3]:= N[ Integrate[Sin[Exp[x]], {x, 0, 1}], 40 ]
Out[3]= 0.8749571987803839994388329808353131568824
```

If you want to evaluate an integral numerically in *Mathematica*, then using **Integrate** and applying **N** to the result is not the most efficient way to do it. It is better instead to use the function **NIntegrate**, which immediately gives a numerical answer, without first trying to get an exact, symbolic, result. You should realize that even when **Integrate** does not in the end manage to give you an exact result, it may spend a lot of time trying to do so.

NIntegrate evaluates numerical integrals directly, without first trying to get a symbolic result.

```
In[4]:= NIntegrate[ Sin[Exp[x]], {x, 0, 1} ]
Out[4]= 0.874957
```

Integrate	NIntegrate	definite integrals
Sum	NSum	sums
Product	NProduct	products
Roots	NRoots	roots of polynomials

Symbolic and numerical versions of some *Mathematica* functions.